

# Session 3.2: Advanced Data Analysis: Feature Engineering, Scaling, and Business Data Preprocessing

PEDAGOGICAL AGENDA

1:2 Delivery Rhythm

## Session Learning Architecture

Structured 3-hour applied computing session designed for business students with zero prior coding experience.

- **Phase 1 (60 Min Lecture):** Computational foundations, business vocabulary, and syntactic mechanics.
- **Phase 2 (120 Min Lab):** Guided hands-on implementation, error diagnosis, and AI co-pilot resolution.
- **Zero Fluff Mandate:** Focus directly on executable Python code, tabular data wrangling, and commercial intelligence.

STRATEGIC OUTCOME

Operational Autonomy

## Business Analytical Competency

Transforming passive spreadsheet users into autonomous programmatic decision makers.

- **Speed & Scale:** Processing datasets exceeding spreadsheet limits with instantaneous vectorized execution.
- **Repeatable Pipelines:** Replacing manual copy-paste workflows with fully automated Python scripts.
- **AI Debugging Literacy:** Leveraging modern IDEs to diagnose and resolve errors independently.

# Core Computational Keywords & Business Meaning

CONCEPT 01

Core Principle

## Syntactic Rigor

Writing clean, self-documenting code that scales across corporate systems.

CONCEPT 02

Core Principle

## Syntactic Rigor

Writing clean, self-documenting code that scales across corporate systems.

CONCEPT 03

Core Principle

## Syntactic Rigor

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CONCEPT 04

Core Principle

## Syntactic Rigor

Writing clean, self-documenting code that scales across corporate systems.

# Computational Foundations & Business Operations

PILLAR 01

Architecture

## Garbage-In, Garbage-Out (The Preprocessing Mandate)

Machine learning algorithms are purely mathematical optimization functions; they cannot ingest raw text strings or missing fields. 80% of enterprise data science time is spent structuring and cleaning features.

PILLAR 02

Architecture

## Dimensionality & Categorical Encoding

When encoding categories with high cardinality, avoid naive label encoding (which implies an artificial mathematical hierarchy like `Master = 2 * Bachelor`). Use One-Hot Encoding (`pd.get_dummies` or `OneHotEncoder`) with `drop_first=True` to prevent multicollinearity.

PILLAR 03

Architecture

## Feature Magnitude & Distance Sensitivity

If `Annual_Income` ranges from `$50,000` to `$200,000` while `Credit_Utilization` ranges from `0.1` to `0.9`, distance-based models (such as K-Nearest Neighbors, Support Vector Machines, and regularized regressions) will prioritize income entirely. Standardization ensures fair feature weighting.

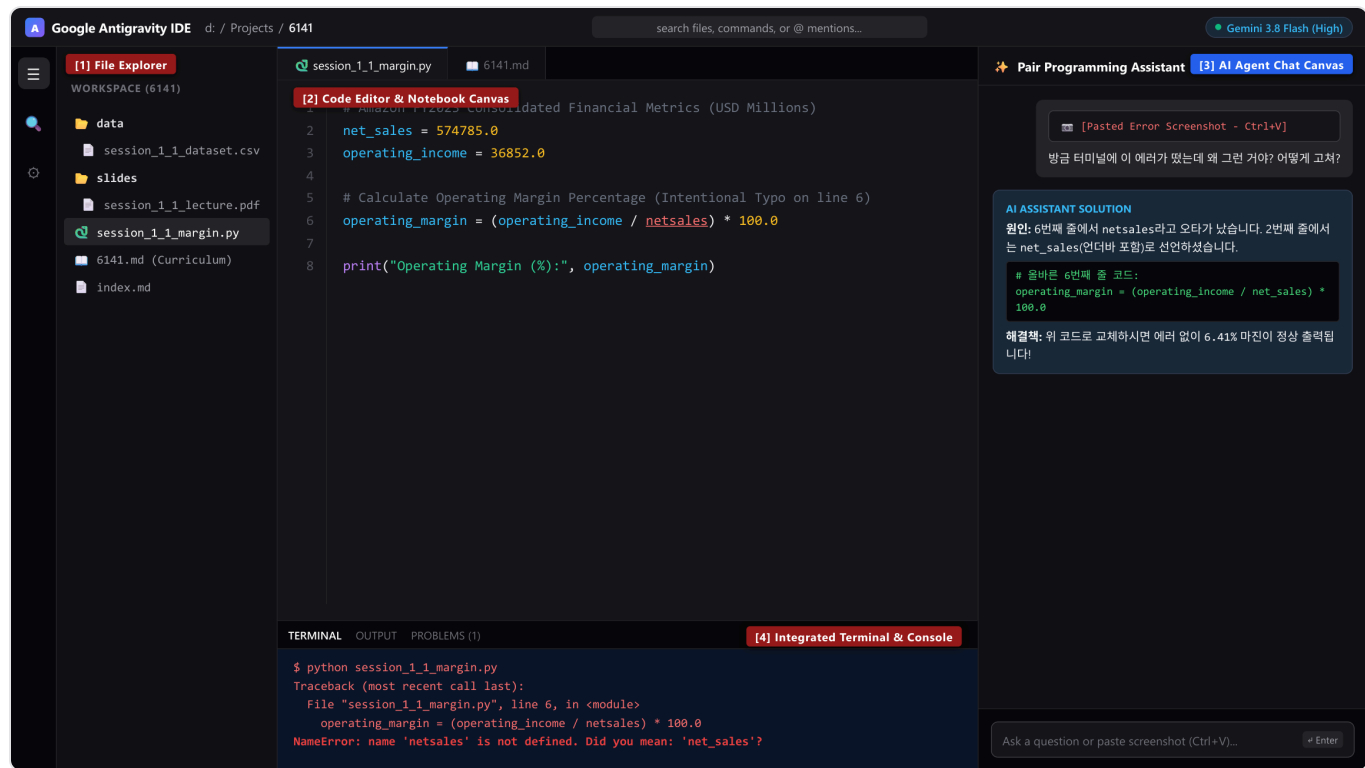
PILLAR 04

Architecture

## Preventing Data Leakage across Train-Test Splits

Never fit standardizers or imputers on the full dataset before splitting. Always split first, fit preprocessing transformers strictly on `X_train`, and transform `X_test` using the learned training parameters.

# Google Antigravity IDE Architecture & AI Debugging Protocol



## [1] File Explorer

### Project Hierarchy

Manage datasets in data/, scripts in root, and exported artifacts.

## [2] Code Editor

### Syntax Workspace

Modern syntax highlighting, lint diagnostics, and line-by-line script authoring.

## [3] AI Agent Chat

### AI Pair Programmer

Paste error screenshots directly to inspect root causes and get verified fixes.

## [4] Terminal

### Execution Engine

Run Python scripts (python script.py) and view console logs and tracebacks.

## Zero-Friction AI Error Resolution Protocol

- Capture:** Press Win + Shift + S to clip terminal error traceback.
- Paste:** Press Ctrl + V into AI Chat [3] with "Explain root cause & fix".
- Apply:** Review the explanation and apply verified fix into Editor [2].

# Script Demonstration 1: Script Demonstration 1

session\_3\_2\_demo\_1.pyPython 3.10

```
1  "kw">class="kw">import pandas "kw">class="kw">as pd
2
3  df = pd."fn">read_csv("data/session_3_2_dataset.csv")
4
5  X_raw = df.drop(columns=["Client_ID", "Defaulted_Binary"])
6  y = df["Defaulted_Binary"]
7  "fn">print("Feature Matrix Shape:", X_raw.shape)
8  "fn">print("Target Vector Distribution:\\n", y.value_counts())
```

\$ python session\_3\_2\_demo\_1.py [Exit 0]

1 Syntactic Structure

Importing production libraries and defining clean analytical data structures.

2 Vectorized Transformation

Applying deterministic operations across the full dataset without fragile cell references.

3 Console Verification

Emitting formatted business KPIs to the terminal for auditability and compliance.

# Script Demonstration 2: Script Demonstration 2

session\_3\_2\_demo\_2.pyPython 3.10

1X\_encoded = pd.get\_dummies(X\_raw, columns=["Education\_Tier"], drop\_first=True, dummies\_na=False)

2 "fn">print("Encoded Feature Columns:\\n", X\_encoded.columns.tolist())

3 "fn">print(X\_encoded."fn">head(2))

\$ python session\_3\_2\_demo\_2.py [Exit 0]

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# Script Demonstration 3: Script Demonstration 3

session\_3\_2\_demo\_3.pyPython 3.10

```
1  "kw">class="kw">from sklearn.model_selection "kw">class="kw">import train_test
2      "kw">class="kw">from sklearn.preprocessing "kw">class="kw">import StandardS
3
4      # Split into 80% Train and 20% Test sets
5      X_train, X_test, y_train, y_test = train_test_split(X_encoded, y, test_size=0.2)
6
7      # Fit scaler strictly on X_train
8      scaler = StandardScaler()
9      X_train_scaled = scaler."fn">fit_transform(X_train)
10     X_test_scaled = scaler."fn">transform(X_test)
11
12     "fn">print(f"X_train scaled shape: {X_train_scaled.shape} | X_test scaled shape: {X_test_scaled.shape}")
```

\$ python session\_3\_2\_demo\_3.py [Exit 0]

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# Interactive Two-Tier Lab Protocol & AI Diagnosis

## EXERCISE 1

Guided Lab

### Foundational Implementation

Step-by-step guided practice establishing baseline syntax mastery in Antigravity IDE.

- **Step 1:** Ingest dataset from `data/session_3_2_dataset.csv`.
- **Step 2:** Implement core analytical functions and compute primary business metrics.
- **Step 3:** Format console outputs and verify calculations against baseline ledger totals.
- **Step 4:** Run in Terminal [4] and confirm clean execution with exit code 0.

## EXERCISE 2+

Coding Challenge

### Advanced Scripting & AI Debugging

Applied programming challenge expanding pipeline logic and resolving intentional exceptions.

- **Task 1 (Pipeline Expansion):** Add secondary business unit logic and multi-tier calculations.
- **Task 2 (Deliberate Error & AI Capture):** Introduce intentional syntax or type error, capture traceback with `Win + Shift + S`, and diagnose via AI Chat [3].
- **Task 3 (Verified Production Run):** Apply verified fix, export audit output, and confirm zero errors with return code 0.